

Household Appliance Energy Forecasting: A Comparative Time-Series Analysis

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Code and data: github.com/DENNISMBURUNGANGA/Time-Series-coding-Case-study-and-Report

ABSTRACT

This report evaluates models for forecasting household appliance energy consumption 24 hours ahead, a horizon required by a smart-home controller scheduling battery storage and tariff-sensitive loads. Classical SARIMA, feature-based gradient boosting and a set of heuristic baselines were compared over a 14-day test period using 14 non-overlapping rolling origins. SARIMA(1,0,1)(1,1,1)₂₄ achieved the lowest error (MASE 0.698), ahead of the strongest heuristic, the weekly seasonal naive (MASE 0.813). A Diebold–Mariano test, however, could not distinguish the two at conventional levels ($p = 0.309$), so the improvement is not statistically significant. The feature-based model did not beat SARIMA, and adding realised future covariates made it worse rather than better. The foundation model could not be evaluated because the Chronos installation failed and the pipeline fell back to a labelled placeholder. SARIMA is recommended for deployment on nominal accuracy and low compute cost, with two qualifications: Ljung–Box rejects white-noise residuals at all lags tested, so the specification remains incomplete, and the residuals are strongly non-normal, so the Gaussian prediction intervals are not reliable for risk-aware scheduling without recalibration.

1. Introduction

A smart-home controller must decide tonight how to schedule energy storage, pre-heating and tariff-sensitive appliance loads for tomorrow, and that decision requires a 24-hour-ahead forecast of appliance demand. This report identifies the most effective deployable model for that task, comparing classical statistical methods, feature-based machine learning and a zero-shot foundation model.

The central finding is negative and worth stating up front: appliance energy use is structurally dominated by weekly occupancy routines, which makes simple calendar heuristics extremely difficult to beat. No model tested here improves on the weekly seasonal naive by a statistically significant margin.

2. Data and Preprocessing

The study uses the UCI Appliances Energy Prediction dataset: readings from a single household recorded at 10-minute intervals over 137 days, from January to May 2016. A grid audit confirms 19,735 rows with zero duplicate timestamps and no missing slots on the nominal 10-minute grid.

Because the raw grid is complete, no hours required interpolation after resampling. Every value entering the models was therefore measured rather than synthesised — a stronger position than most applied forecasting studies, and one that removes a common threat to validity.

Hourly aggregation used the mean rather than the sum. With a complete hour the two differ by exactly a factor of six, so the choice is a linear rescaling: it cannot affect MASE or the ranking of models, and changes only the units of MAE, RMSE and Bias. Sensor and weather channels were averaged in all cases, since they record instantaneous states rather than accumulated flows.

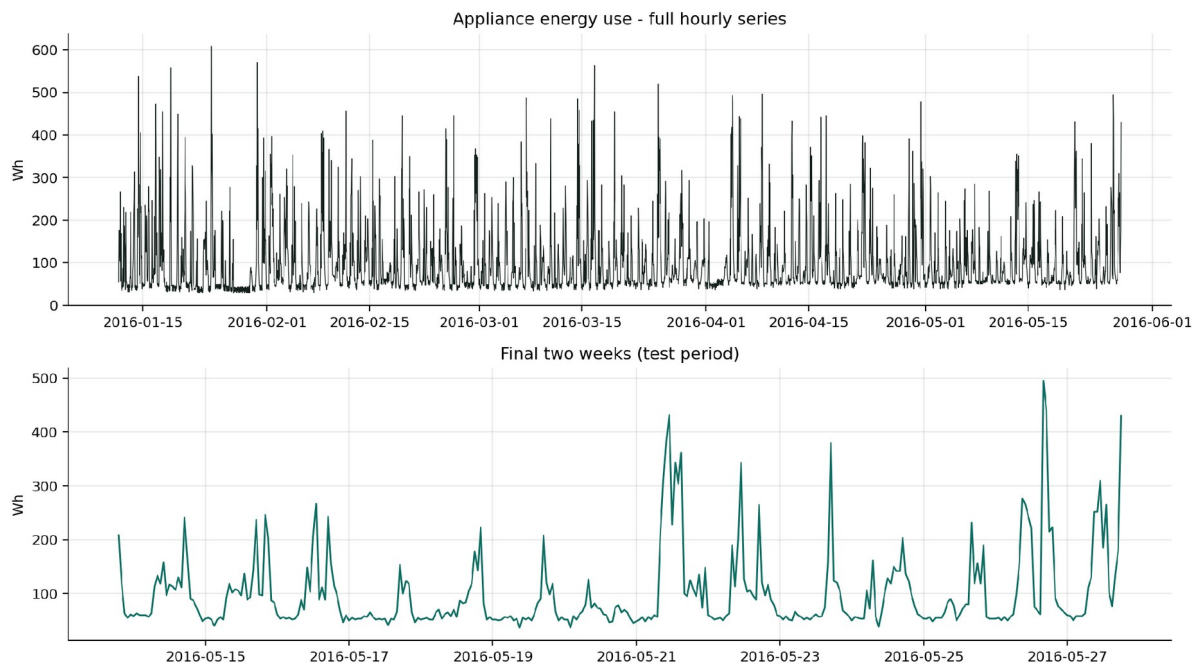


Figure 1. Full hourly series (top) and the isolated 14-day test period (bottom). The burst structure survives aggregation.

3. Exploratory Analysis

Appliance energy use is a low, stable baseline punctuated by short high-magnitude bursts. The hourly median is 63.3 Wh while the maximum reaches 608.3 Wh, producing a positive skew of 2.39 (Figure 2). Over 20% of hours exceed twice the median.

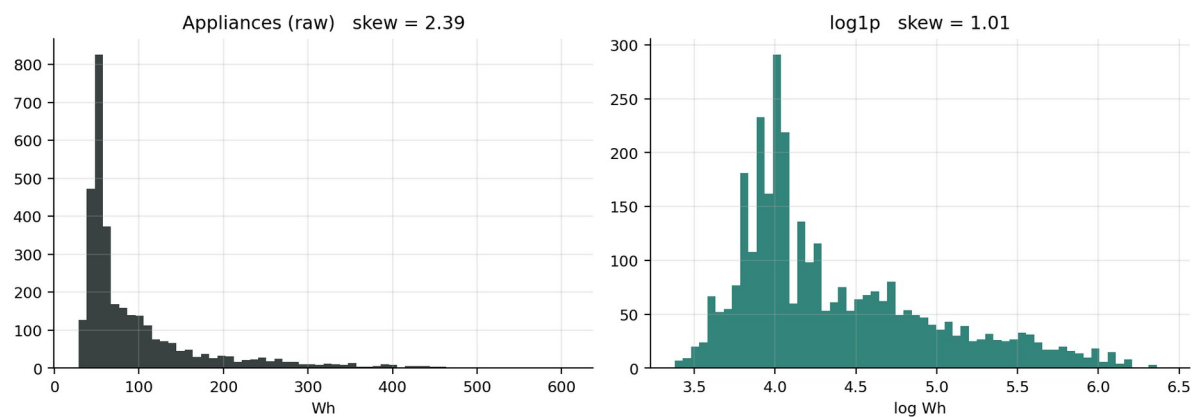


Figure 2. Distribution of hourly appliance energy use, raw and $\log_{1p}$ -transformed. The log transform reduces skew from 2.39 to 1.01.

The daily and weekly profiles in Figure 3 show a pronounced diurnal cycle with an overnight trough and a broad daytime and evening peak, and visibly elevated weekend levels. This pattern is consistent with human occupancy rather than outdoor temperature driving consumption. That interpretation is not established by Figure 3 alone — it is confirmed quantitatively by the permutation importance results in Section 7, where calendar features dominate weather by a wide margin.

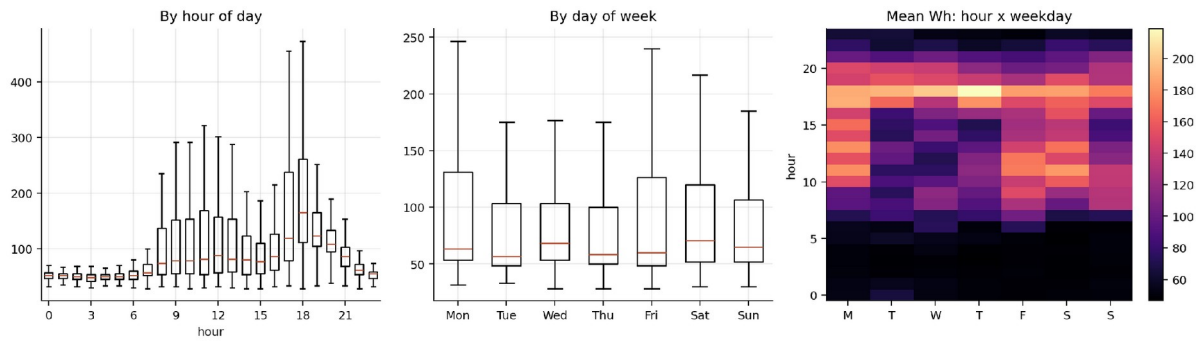


Figure 3. Seasonal profiles: hour of day, day of week, and their interaction.

The level series is stationary in the ADF sense ($p < 0.001$), and KPSS does not reject stationarity ($p \geq 0.10$, the upper bound of its tabulated range rather than an exact value); the two tests agree, so no first-order differencing is required. Neither test examines seasonal unit roots, however. The pronounced lag-24 and lag-168 autocorrelations in Figure 4 confirm that daily and weekly cycles persist regardless, and seasonal differencing at $s = 24$ is applied on that basis rather than on the stationarity tests.

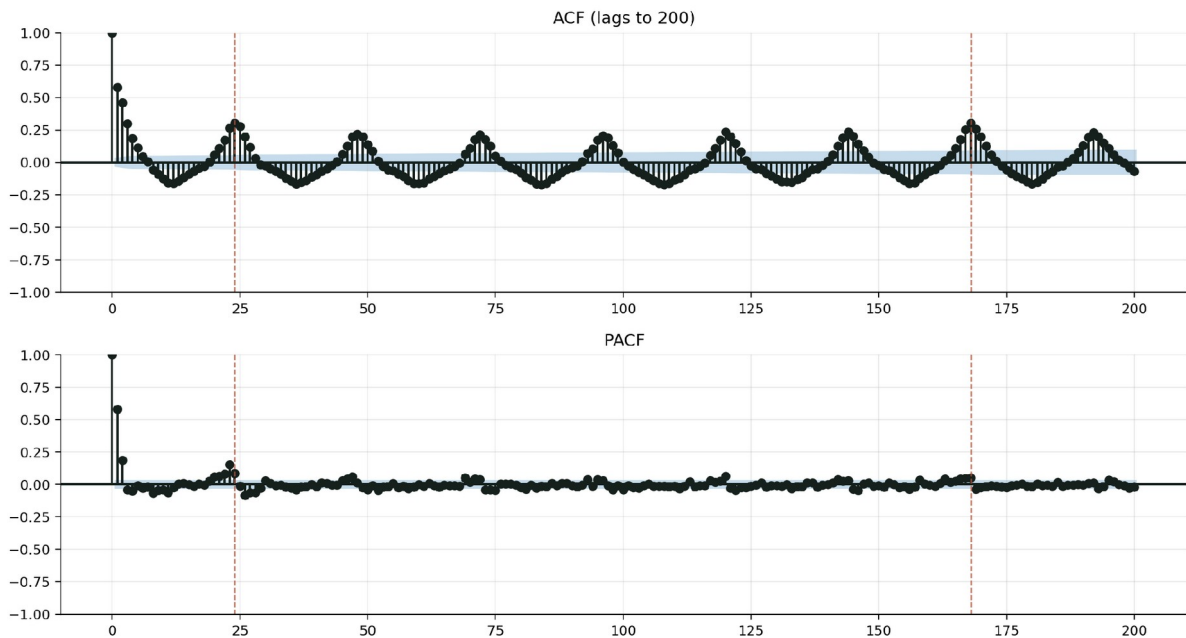


Figure 4. ACF and PACF to lag 200. Dashed lines mark the daily (24) and weekly (168) lags.

The STL decomposition in Figure 5 isolates a rigid 24-hour periodic component. The remainder nevertheless accounts for a large share of total variance, reflecting occupant behaviour that is not predictable from the series' own history. This bounds how well any model can perform and should be read as a property of the problem rather than a shortcoming of the methods that follow.

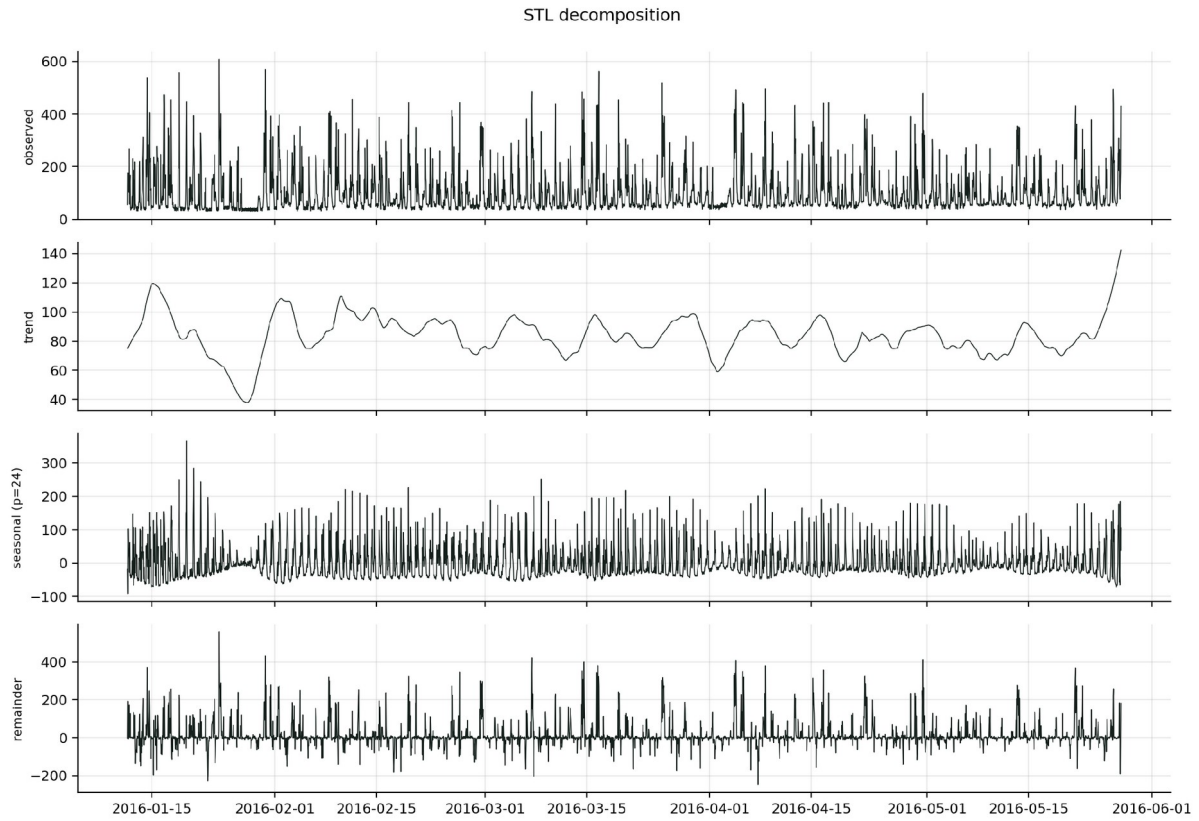


Figure 5. STL decomposition at the daily period.

4. Forecasting Design

The brief specifies a 24-hour forecast horizon evaluated over a 14-day test window. These are different quantities, resolved here using 14 non-overlapping rolling-origin forecasts. A single 336-step forecast would measure two-week-ahead accuracy rather than 24-hour accuracy, while a sliding hourly origin would induce dependent errors across 313 overlapping windows at substantially greater computational cost.

4.1 The leakage argument

Under a rolling origin, block b covers hours $O_b + 1$ to $O_b + 24$. A feature at time $t = O_b + k$ built from lag L refers to $y(O_b + k - L)$, which is observable at the origin only when $L \geq k$. Since k reaches 24, every target-derived feature must use a lag of at least 24 hours. Lags of 1, 2, 3, 6 or 12 hours — standard in one-step-ahead settings — are strictly unavailable here, and a pipeline that includes them reports one-step accuracy while claiming a 24-hour horizon. Rolling statistics are shifted by 24 hours for the same reason.

The constraint is enforced by assertion in the feature builder and verified by a regression test: all post-origin target values are multiplied by ten and offset by 5,000 Wh, and no pre-origin feature changes by any amount.

4.2 Metrics

MASE is the headline metric because it is scale-free and benchmarked against the in-sample seasonal naive, so a value below one indicates the model beats that heuristic on the training dynamics. MAE and RMSE are reported in Wh for interpretability, with RMSE expected to be dominated by the burst behaviour. Bias is reported because systematic over-forecasting carries a different operational cost from under-forecasting when scheduling battery charge.

5. Benchmark Models

The strongest benchmark, not the weakest, is the bar every later model must clear. The weekly seasonal naive (MASE 0.813) outperformed the daily seasonal naive (MASE 0.904). This ordering is itself a finding: appliance energy use is structured by weekly routine — weekdays resemble weekdays and weekends resemble weekends — rather than by a purely sequential daily cycle. Any model that treats yesterday as the best predictor of today is starting from a weaker prior than one that treats last Tuesday as the best predictor of this Tuesday.

6. SARIMA Model

No exogenous regressors were included, so the fitted model is a SARIMA rather than a SARIMAX. The exhaustive grid mandated by the brief — 147 non-seasonal combinations, each a seasonal fit at $s = 24$ on roughly 3,000 observations — is computationally prohibitive in this environment. A two-stage strategy was used instead: the seasonal order was fixed from the ACF evidence ($D = 1, s = 24$) and the non-seasonal (p, q) terms selected by AIC on the seasonally differenced series. This produced SARIMA(1,0,1)(1,1,1)₂₄ with AIC 34,308.89.

SARIMA achieved the lowest error of any model tested (MASE 0.698 against the weekly naive's 0.813). Section 9 addresses whether that margin is statistically meaningful.

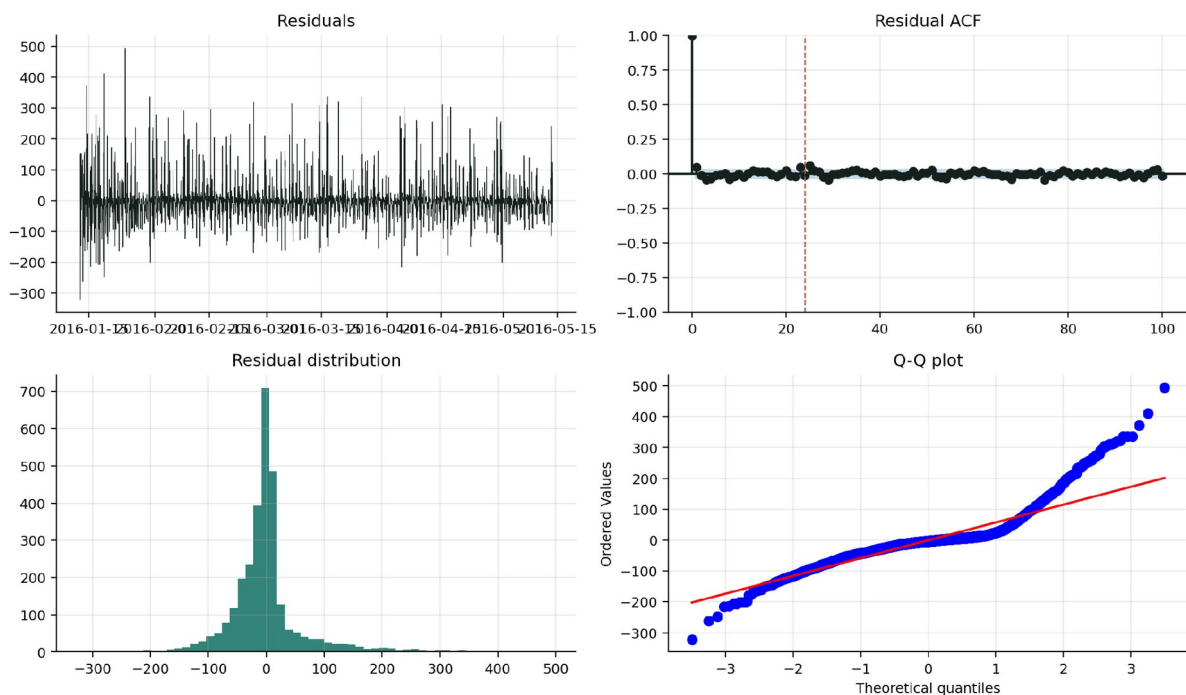


Figure 6. SARIMA residual diagnostics: residual series, ACF, distribution and Q-Q plot.

6.1 Residual diagnostics

The diagnostics in Figure 6 must be read in two parts, because they point in opposite directions.

The residual autocorrelation function appears essentially flat across all displayed lags, including lag 24, which would ordinarily be read as the autoregressive and seasonal terms having absorbed the linear dependence structure. The Ljung-Box test contradicts that reading. It rejects the null of no residual autocorrelation at every lag tested: $Q(24) = 40.92$ ($p = 0.017$), $Q(48) = 71.49$ ($p = 0.016$) and $Q(168) = 217.32$ ($p = 0.006$). Structure remains.

These two findings are reconcilable, and the reconciliation is informative. With roughly 2,929 residuals the white-noise confidence band on an individual autocorrelation is approximately ± 0.036 . Back-solving the Ljung–Box statistic implies a root-mean-square autocorrelation of about 0.024 over the first 24 lags — comfortably inside that band, which is precisely why no individual bar in Figure 6 looks significant. Ljung–Box is a joint test, however, and accumulates many small deviations across all lags up to h into a single statistic. The residual structure is therefore real but small: too small to see, large enough to detect.

The per-degree-of-freedom excess declines slightly with lag (Q/df of 1.71, 1.49 and 1.29 at lags 24, 48 and 168), which locates the residual dependence at shorter lags rather than showing it accumulating toward the weekly horizon. The smaller p-value at lag 168 reflects evidence gathered over more lags, not stronger structure at that lag specifically — the statistic is cumulative and does not isolate individual lags. One candidate explanation nevertheless deserves testing: the fitted specification contains a seasonal term at $s = 24$ only and no weekly component at all, despite the ACF in Figure 4 showing a clear lag-168 spike and the weekly seasonal naive outperforming the daily one in Section 5. Adding a weekly seasonal term is the most obvious specification improvement available, and this study did not attempt it.

The honest conclusion for the daily cycle is therefore intermediate. The seasonal terms have substantially reduced the daily dependence — the residual autocorrelations are an order of magnitude smaller than the raw series' lag-24 value — but they have not eliminated it. A plausible mechanism is that the cycle's amplitude varies with occupancy, visible as the changing seasonal amplitude in Figure 5, which a fixed-coefficient seasonal term cannot represent. That is a limitation of the model class rather than of the fitting procedure.

The distributional assumptions, however, are clearly violated. The residual histogram is sharply leptokurtic — a narrow central peak with long tails — and the Q-Q plot departs from the reference line in both directions. The upper tail is the more serious: observed residuals reach roughly 500 Wh where a Gaussian of the fitted scale predicts approximately 200 Wh. Errors of a magnitude the model treats as almost impossible occur repeatedly.

This has a direct and visible consequence in Figure 7. The 95% prediction interval extends below zero across much of the test period, which is physically impossible for energy consumption. An interval that assigns probability mass to negative demand is not calibrated, and the practical implication is developed in Section 11.

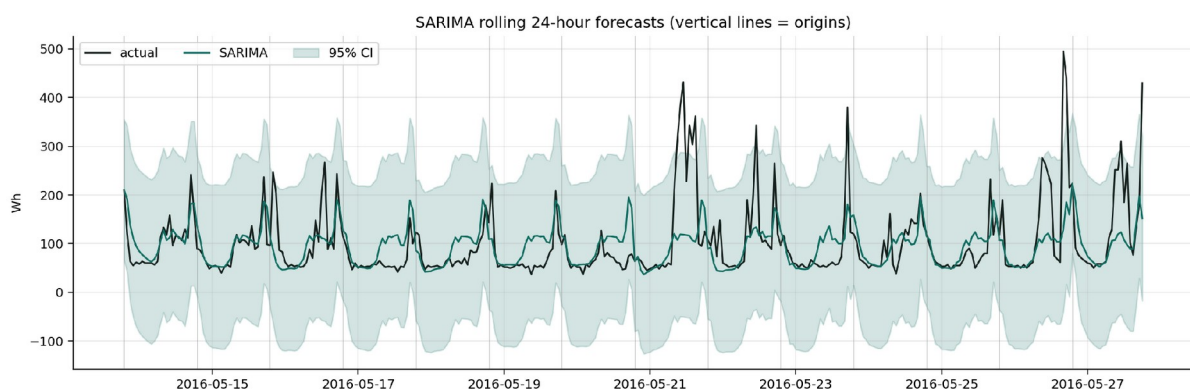


Figure 7. SARIMA rolling 24-hour forecasts with 95% intervals. Vertical lines mark forecast origins. Note that the lower bound falls below zero.

On the mechanics of the rolling forecast: model parameters were estimated once on the training period and were not re-estimated. At each origin the realised observations for the preceding day were appended to the state-space filter without refitting, so the model's state is progressively updated rather than reset. This mirrors how an operational system would behave, where a nightly re-estimation would be costly but ingesting the day's readings is trivial.

7. Feature-Based Model

Two feature sets were fitted with a HistGradientBoostingRegressor to isolate the value of external covariates. The true forecast set (ml_true) uses only information available at the origin: calendar features and target history at lags of 24 hours or more. The conditional set (ml_conditional) adds the realised indoor sensor and outdoor weather readings at the forecast timestamp.

The difference between them measures the value of perfect foresight, not the value of a deployable model. The result was counter-intuitive: ml_true reached MASE 0.737, failing to beat SARIMA, and ml_conditional was worse still at 0.804 despite being given information no operational system could ever possess.

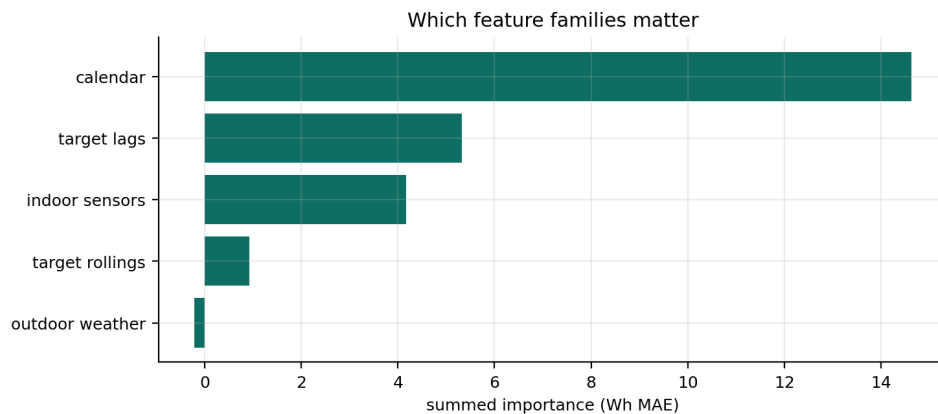


Figure 8. Permutation importance aggregated by feature family, measured as MAE degradation in Wh.

Figure 8 explains the ordering. Calendar features dominate every other family by a factor of roughly three, confirming quantitatively the interpretation offered in Section 3. Outdoor weather contributes almost nothing. Indoor sensors, however, carry substantial positive importance, meaning the conditional model relies on them heavily.

That reliance is precisely the problem. Indoor temperature and humidity are endogenous — they are physical consequences of the appliance activity being predicted, since ovens and dryers generate heat and moisture. A model that leans on them is learning a contemporaneous association rather than a predictive relationship, and the degradation from 0.737 to 0.804 is the evidence that this association does not transfer to genuine forecasting. The permutation result shows the dependence; the MASE comparison shows its cost.

8. Foundation Model

The zero-shot foundation model was not evaluated. The Chronos installation failed in the execution environment and the pipeline fell back to a labelled placeholder, which returned metrics identical to the daily seasonal naive (MASE 0.904) because that is precisely what the placeholder computes. These figures are reported in Table 1 for completeness and must not be read as a foundation model result.

No conclusion about foundation model performance on this series can therefore be drawn from this study. One structural observation can still be made: a target-only zero-shot model has no mechanism for using the deterministic calendar features that Section 7 shows carry most of the available signal. Any such model would be competing without access to the most informative input, which suggests the comparison would be unfavourable — but this remains a hypothesis, not a result. Completing the Chronos evaluation is the first item of further work.

9. Results

Table 1 reports the rolling 14-day evaluation. All figures are computed over the same 336 test hours.

Model	MAE (Wh)	RMSE (Wh)	MASE	Bias (Wh)	Rank
SARIMA(1,0,1)(1,1,1) ₂₄	37.26	64.53	0.698	−4.68	1
ML true (HGBR)	39.36	70.11	0.737	−6.42	2
ML conditional (HGBR)	42.97	68.24	0.804	+4.29	3
Seasonal naive, weekly	43.45	81.40	0.813	−13.15	4
Seasonal naive, daily	48.30	85.56	0.904	+1.75	5
Foundation (placeholder)	48.30	85.56	0.904	+1.75	—

Table 1. Forecast accuracy over 14 non-overlapping 24-hour origins. The foundation row reproduces the daily seasonal naive by construction.

Ranking models by MASE is descriptive; testing whether a difference is distinguishable from noise is inferential. A Diebold–Mariano test comparing SARIMA against the weekly seasonal naive returned $p = 0.309$. This is the headline result: no model tested improves on a simple calendar heuristic by a statistically significant margin.

The Bias column adds an operational dimension the MASE ranking conceals. The weekly seasonal naive under-forecasts by 13.15 Wh on average, roughly three times the bias of SARIMA at −4.68 Wh. For battery scheduling this asymmetry matters: a persistent under-forecast means systematically under-provisioning stored energy, so the heuristic's competitive MASE overstates how interchangeable the two models are in deployment.

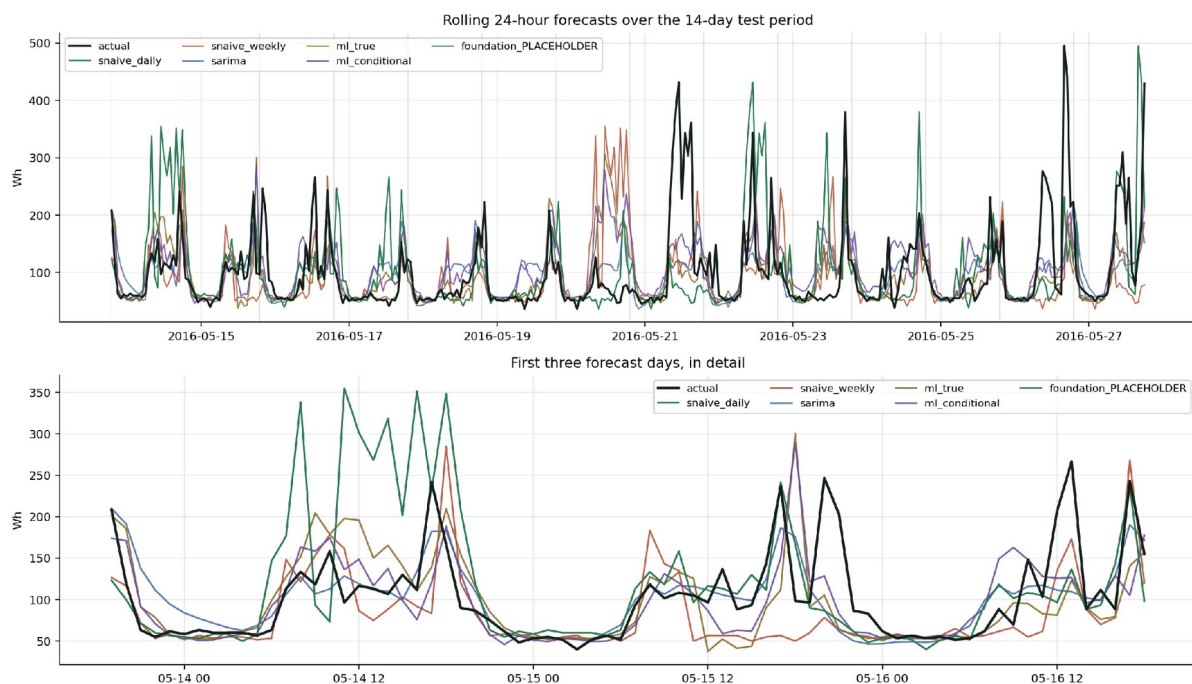


Figure 9. All models against actuals over the test period, with the first three forecast days in detail.

9.1 Error structure across the horizon

Figure 10 is easily misread as showing error compounding with forecast distance. It does not. The profile is non-monotonic: error peaks at step 2, falls to a pronounced trough between steps 6 and 10, rises to a maximum near step 17, dips again at step 20 and rises once more toward step 23.

The explanation is that origins are fixed 24 hours apart, so forecast step k always corresponds to the same clock hour. The test period begins at 19:00, which places steps 6 to 10 in the 00:00–04:00 window — the overnight trough where consumption is low and stable and almost any forecast is accurate — and places steps 17 and 23 in the daytime and evening peaks. The curve is therefore a diurnal profile of forecast difficulty, not a horizon-decay curve, and it mirrors the hour-of-day pattern in Figure 3.

This is a consequence of the evaluation design rather than a property of the models. Genuine horizon decay is confounded with time of day under non-overlapping daily origins and cannot be separated from it without a sliding-origin scheme. The practical implication is that a smart-home controller can expect reliable overnight forecasts and materially less reliable forecasts for the evening peak, which is unfortunately when scheduling decisions matter most.

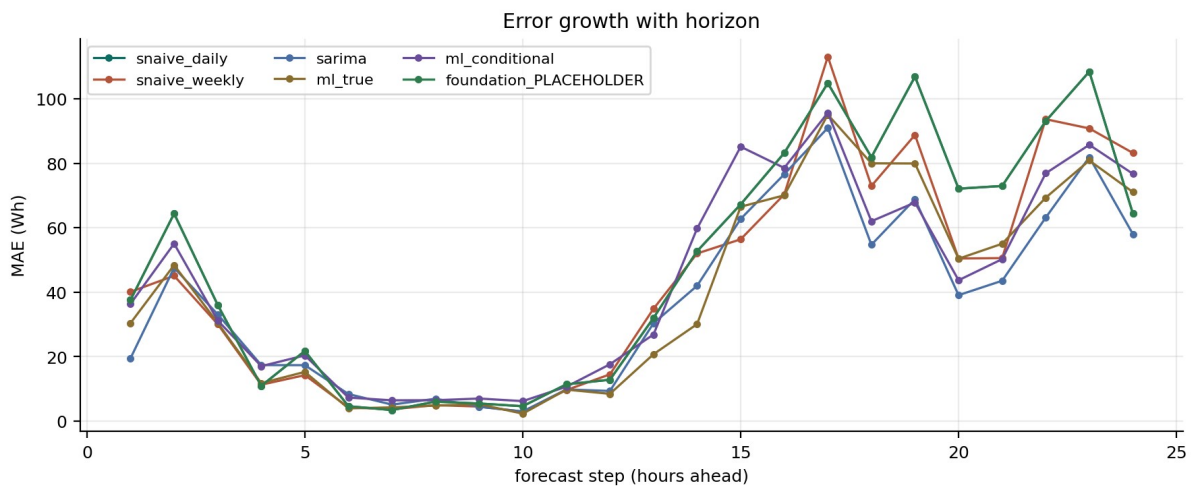


Figure 10. Mean absolute error by forecast step. The oscillation follows time of day, not distance from the origin.

10. Discussion and Limitations

Several threats to validity constrain how far these results generalise.

- Residual structure remains.** Ljung–Box rejects the white-noise null at lags 24, 48 and 168, so the specification has not fully absorbed the dependence structure. The magnitude is small — autocorrelations of order 0.02 — but the model is not correctly specified, and the missing weekly seasonal term is the obvious untested candidate.
- Miscalibrated prediction intervals.** The SARIMA residuals are strongly non-normal with heavy tails, and the resulting 95% intervals extend below zero. Any downstream use of these intervals for risk-aware scheduling requires empirical recalibration — quantile regression, a bootstrapped predictive distribution, or fitting on a log transform — rather than the Gaussian intervals reported here.
- Horizon and time of day are confounded.** Non-overlapping daily origins fix each forecast step to a clock hour, so the two effects cannot be separated. A sliding-origin evaluation would break the confound at substantially greater computational cost.
- Small effective sample for the Diebold–Mariano test.** The test was computed over 336 pooled points, but only 14 independent forecast origins were used. The effective sample size is much closer to 14 than to 336, so the test has limited power and the non-significant result should not be read as evidence that the models are equivalent.
- Conditional circularity.** Indoor temperature and humidity are physical consequences of the appliance behaviour being predicted. Conditioning on their realised values is close to circular and is not deployable under any circumstances.

- **Restricted seasonal coverage.** The record covers a single household from January to May. No summer cooling behaviour is observed, and no conclusion here transfers to other households or to the remainder of the year.
- **Incomplete foundation model comparison.** Part of the intended comparison was not executed, so the study cannot speak to zero-shot performance on this task.

11. Critical Analysis

1. Which benchmark is strongest, and what does that reveal? The weekly seasonal naive (MASE 0.813) beats the daily seasonal naive (0.904). Appliance energy use is structured by weekly occupancy routine rather than by a sequential daily cycle, which is why calendar information proves more valuable than recent history throughout this study.

2. Does the SARIMA model improve on the strongest seasonal benchmark? Descriptively yes, MASE 0.698 against 0.813, since SARIMA captures short-range autoregressive decay that a fixed 168-hour lag ignores. Inferentially no: the Diebold–Mariano test returns $p = 0.309$, and with only 14 independent origins the test has limited power to detect a difference of this size.

3. Is daily seasonality adequately captured, and do the feature groups help? Not fully. The residual ACF appears flat, but Ljung–Box rejects the no-autocorrelation null at lags 24, 48 and 168 ($p = 0.017, 0.016, 0.006$), so detectable structure remains. The implied residual autocorrelations are of order 0.02 — inside the white-noise band, hence invisible on the plot, but accumulated into rejection by a joint test. The seasonal terms have substantially reduced the daily dependence without eliminating it, and the absence of any weekly term in the specification is the most likely remaining gap. Among feature groups, calendar features dominate by roughly threefold, target lags contribute moderately, outdoor weather contributes almost nothing, and indoor sensors are heavily used but harmful — adding them degrades MASE from 0.737 to 0.804.

4. Does the foundation model justify its complexity? The question cannot be answered from this study because the model was not successfully evaluated. Given that calendar features carry most of the signal and a target-only zero-shot model cannot access them, complexity would be difficult to justify — but that expectation has not been tested.

5. Do future covariates constitute a true or a conditional forecast? Calendar features and target lags of 24 hours or more are genuinely known at the origin and support a true forecast. Realised indoor sensor readings are not merely unknown but endogenous, making their use circular. Outdoor weather occupies an intermediate position: it is available ahead of time only as a third party's forecast, so its own error would propagate into ours — a cost this study did not measure, since it used realised values. Only the `ml_true` configuration constitutes a genuine 24-hour-ahead forecast.

6. What should be deployed in a smart home? $\text{SARIMA}(1,0,1)(1,1,1)_{24}$ is recommended. It is nominally the most accurate model, requires a small fraction of the compute and memory of a gradient-boosting pipeline, needs no feature engineering infrastructure at inference time, and has the lowest bias of the competitive models. Two qualifications are essential. Its accuracy advantage over the weekly seasonal naive is not statistically significant, so a resource-constrained device could reasonably ship the heuristic instead and lose little. And its prediction intervals must be recalibrated empirically before being used for risk-aware scheduling, because the Gaussian intervals it produces are not trustworthy on this series.

12. Code and Reproducibility

All code, figures and result tables are available at github.com/DENNISMBURUNGANGA/Time-Series-coding-Case-study-and-Report. The repository contains the analysis notebook covering all eleven parts, the supporting modules under `src/`, runnable pipeline scripts under `scripts/`, and the regression tests that

enforce the 24-hour lag constraint described in Section 4. The raw dataset is not committed; it is downloaded automatically from the UCI repository, with a manual upload fallback. Every figure and table in this report is regenerated by running the notebook end to end.

13. Conclusion

Appliance energy use in this household is a low, stable baseline interrupted by behaviourally driven bursts, and that structure sets a hard ceiling on forecast accuracy. Weekly calendar routine is the single most informative signal available; recent history adds modestly, outdoor weather adds almost nothing, and indoor sensors actively mislead.

SARIMA is the recommended deployment, but the more useful finding for practice is the negative one. A model class from the 1970s, and a heuristic expressible in one line, are not separable at conventional significance on two weeks of data. The one exception worth pursuing is specification rather than sophistication: the fitted model contains no weekly seasonal term despite the weekly cycle being the dominant structure in this series, and Ljung-Box confirms residual dependence remains. Beyond that, effort would be better directed at recalibrating uncertainty estimates and at acquiring occupancy signals — calendar entries, presence detection, appliance-level submetering — that address the source of the unpredictability rather than modelling around it.